

Viscoplastic Squeeze

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Context and problematic. A viscoplastic squeeze flow is triggered when a viscoplastic fluid is compressed between two approaching surfaces. These materials tend to behave either like non-deformable (ideal) solids at low (compressive) stresses or like complex liquids at higher stresses. Although the viscoplastic squeeze dynamics is directly related to numerous applications (such as the assembling of electronics plates used in health-devoted devices by compressing an adhesive fluid between them, compression moulding, papermaking, and the conception of rheometry devices), the physical mechanisms driving the flow are still poorly understood [1-4]. The lack of knowledge in the field becomes even more critical when these materials spread upon rough surfaces.

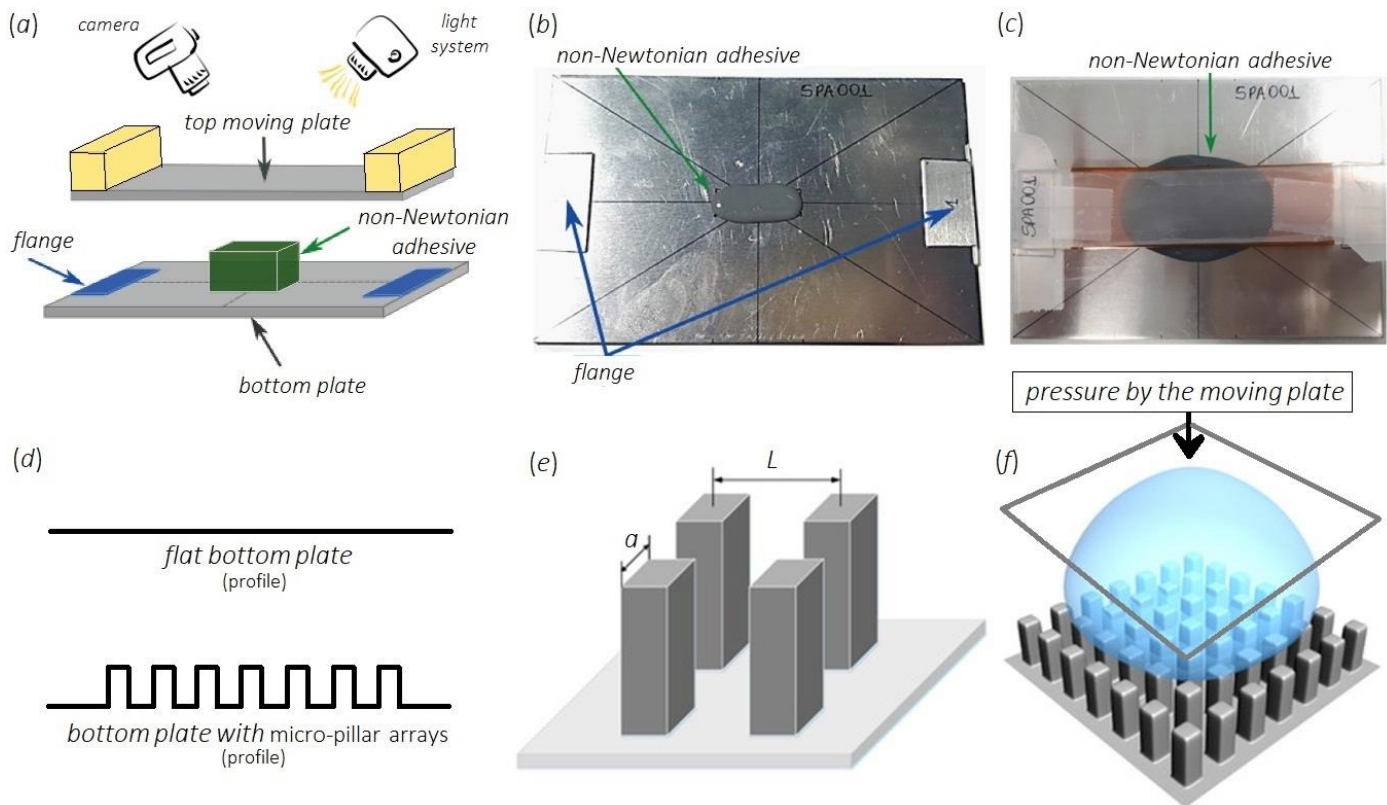


Figure 1. (a-c) Viscoplastic fluid (green) squeezed between parallel flat plates [adapted from 4]. (d) Two plate profiles with and without micro-pillar arrays. (e) Micro-pillars morphology. (f) Viscoplastic fluid (in blue) squeezed between micro-pillar arrays.

Objective and method. This project investigates the **physical mechanisms** driving the spreading dynamics (e.g., spreading against time) of a viscoplastic adhesive (in green in figures 1a-c) squeezed by two parallel plates with and without micro-pillar arrays (see the plate profiles in figures 1d-f), i.e., **viscoplastic squeeze** [1-3]. The analyses are conducted through a **mixed approach combining experiments and numerical simulations**. Experiments are performed using a commercial fluid (3M epoxy adhesive enriched with gold-coated polymeric particles), as well as using Carbopol gels (a well-known yield-stress fluid model). The compression process and the subsequent fluid spreading are followed using a high-speed camera. The numerical simulations are based on an adaptive variational multi-scale method for two fluids (non-Newtonian phase and air). The physical mechanisms driving the analysed phenomenon are underlined through dynamic analyses, energy budgets, and scaling laws [3].

Keywords: squeeze; viscoplastic fluid; experiments; numerical simulations

Reference

- [1] Engmann et al., [Squeeze flow theory and applications to rheometry: A review](#), *JNNFM*, 2005
- [2] Koblitz et al., [Viscoplastic squeeze flow between two identical infinite circular cylinders](#), *Physical Review Fluids*, 2013
- [3] Isukwem & Pereira, [Dam break of viscoplastic elliptical objects](#), *JNNFM*, 2025
- [4] Silva et al., [Numerical simulations of adhesive spreading during bonding-induced squeeze](#), *OATAO*, 2021